



University Institute of Engineering
Department of Computer Science & Engineering

Experiment:8

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Subject Name: DBMS

1. Aim of the practical: T

o create and execute a stored procedure that retrieves employee data and calculates bonus dynamically using input parameters, demonstrating the use of functions and procedural logic in PostgreSQL.

2. Tool Used:

● **Database Management System:**

○ **PostgreSQL**

● **Database Administration Tool:**

○ **pgAdmin**

3. Objective:

To write and execute a PL/SQL stored procedure that dynamically accepts gender as an argument and computes the employee count corresponding to the given gender.



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4. Practical / Experimental Steps

Step 1: Create the base table employees with attributes such as emp_id, emp_name, salary, and department.

Step 2: Insert sample records into the employees table to simulate real-world employee data.

Step 3: Define and create a stored procedure (function) GET_BONUS that accepts employee ID as an input parameter.

Step 4: Implement logic inside the procedure to retrieve the employee's salary and calculate the bonus using a percentage formula.

Step 5: Execute the stored procedure by passing dynamic input values using a DO block.

Step 6: Display the output using RAISE NOTICE to show employee ID and calculated bonus.

Step 7: Verify the correctness of the output and analyze how the procedure handles dynamic inputs and returns computed results.

5. I / O Analysis

A) Create Table

```
CREATE TABLE employees (  
    emp_id INT PRIMARY KEY,  
    emp_name VARCHAR(100) NOT NULL,  
    salary NUMERIC(10,2) NOT NULL,  
    department VARCHAR(50)  
);
```

```
CREATE TABLE  
  
Query returned successfully in 138 msec.
```



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B) Insert Sample Data

```
INSERT INTO employees(emp_id,emp_name, salary, department) VALUES  
(1,'Alice', 50000, 'HR'),  
(2,'Bob', 60000, 'IT'),  
(3,'Charlie', 55000, 'Finance');
```

```
INSERT 0 3
```

```
Query returned successfully in 62 msec.
```

C) Functions and Their Output

```
CREATE OR REPLACE PROCEDURE get_Employee_Count_BY_Gender(IN P_GEN VARCHAR(20), OUT  
COUNT_EMP INT, INOUT STATUS VARCHAR)  
AS
```

```
$$
```

```
BEGIN
```

```
SELECT COUNT(*) INTO COUNT_EMP FROM EMPLOYEES WHERE GENDER=P_GEN;  
STATUS:='SUCCESS';
```

```
END;
```

```
$$ LANGUAGE PLPGSQL
```

```
DO
```

```
$$
```

```
DECLARE
```

```
GEN VARCHAR(20):='Male';
```

```
COUNT_BY_GEN INT;
```

```
STATUS VARCHAR(20):='Fail';
```

```
BEGIN
```

```
CALL get_Employee_Count_BY_Gender(GEN,COUNT_BY_GEN,STATUS);
```

```
RAISE NOTICE 'GENDER IS % YOUR COUNT IS % AND STATUS IS  
%',GEN,COUNT_BY_GEN,STATUS;
```

```
END;
```

```
$$
```

```
NOTICE: GENDER IS Male YOUR COUNT IS 2 AND STATUS IS SUCCESS  
DO
```

```
Query returned successfully in 68 msec.
```

```
CREATE OR REPLACE FUNCTION Get_Data()  
RETURNS TABLE(name_emp VARCHAR(10), emp_salary NUMERIC(10,2))  
AS
```



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```
$$  
BEGIN  
RETURN QUERY SELECT emp_name, salary FROM employees;  
END;  
$$ LANGUAGE PLPGSQL;
```

```
CREATE FUNCTION
```

```
Query returned successfully in 60 msec.
```

```
SELECT * FROM Get_Data();
```

	name_emp character varying	emp_salary numeric
1	Amit	30000.00
2	Riya	35000.00
3	John	28000.00
4	Sneha	40000.00

6. Learning outcomes (What I have learnt):

- Understand the structure and components of a stored procedure.
- Implement IN and OUT parameters in PL/SQL procedures.
- Pass dynamic input values to stored procedures.
- Apply aggregate functions within procedures for data analysis.
- Develop reusable database logic for real-world business scenarios used in companies

like CData and Toddle.